

Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.7)

Metric

$$ds^2 = (-x^2(2A_x(u) + B(u)) + y^2(2A_x(u) - B(u)) + 4xyA_y(u)) du \otimes du - du \otimes dv - dv \otimes du + dx \otimes dx + dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(x^2 A_x(u) - y^2 A_x(u) - 2xyA_y(u) + \frac{1}{2}x^2 B(u) + \frac{1}{2}y^2 B(u) \right) du + (1) dv$$

$$m = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(\frac{1}{2} i \sqrt{2} \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(-\frac{1}{2} i \sqrt{2} \right) dy$$

Contravariant components

$$l = (-1) dv$$

$$n = (-1) du + \left(x^2 A_x(u) - y^2 A_x(u) - 2xyA_y(u) + \frac{1}{2}x^2 B(u) + \frac{1}{2}y^2 B(u) \right) dv$$

$$m = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(\frac{1}{2} i \sqrt{2} \right) dy$$

$$\overline{m} = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(-\frac{1}{2} i \sqrt{2} \right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u + x^2 A_x(u) - y^2 A_x(u) - 2xy A_y(u) + \frac{1}{2} x^2 B(u) + \frac{1}{2} y^2 B(u) \partial_v$$

$$\delta = \frac{1}{2} \sqrt{2} \partial_x + \frac{1}{2} i \sqrt{2} \partial_y$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2} \partial_x - \frac{1}{2} i \sqrt{2} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = \frac{1}{2} \sqrt{2} (2x A_x(u) + 2i y A_x(u) + 2i x A_y(u) - 2y A_y(u) + x B(u) - i y B(u))$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = B\left(u\right)$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = 2\,A_x\left(u\right) + 2i\,A_y\left(u\right)$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "N"